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Assignment 8

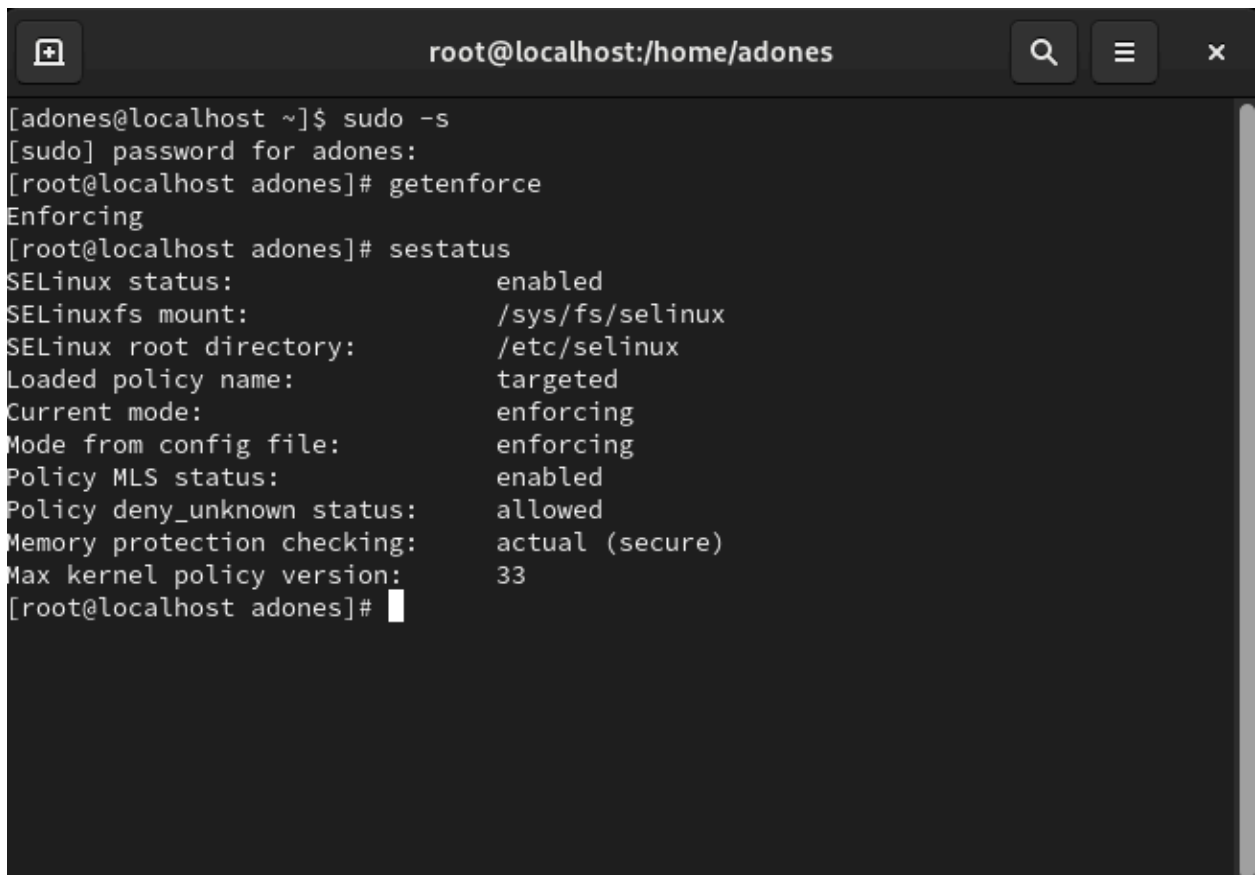
Assignment Overview:

The goal of this assignment is to research SELinux. This task involves creating a virtual machine to get hands-on experience using SELinux to get familiar with some of the common commands. The final deliverable will be a report of your findings including screenshots of your results from the contexts and labels you reviewed. Further details are included below.

Main Tasks:

This project will encompass the following tasks:

1. Create a virtual machine for a RHEL-based OS
2. Review and confirm SELinux is enabled and enforcing

A terminal window titled 'root@localhost:/home/adones' with search, menu, and close icons. The terminal shows a user 'adones' running 'sudo -s' to become root. Then, 'getenforce' is run, showing 'Enforcing'. Finally, 'sestatus' is run, displaying SELinux status: enabled, SELinuxfs mount: /sys/fs/selinux, SELinux root directory: /etc/selinux, Loaded policy name: targeted, Current mode: enforcing, Mode from config file: enforcing, Policy MLS status: enabled, Policy deny_unknown status: allowed, Memory protection checking: actual (secure), and Max kernel policy version: 33.

```
root@localhost:/home/adones
[adones@localhost ~]$ sudo -s
[sudo] password for adones:
[root@localhost adones]# getenforce
Enforcing
[root@localhost adones]# sestatus
SELinux status:                enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:              targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Memory protection checking:     actual (secure)
Max kernel policy version:      33
[root@localhost adones]#
```

```
root@localhost:/home/adones

# This file controls the state of SELinux on the system.
# SELINUX= can take one of these three values:
#   enforcing - SELinux security policy is enforced.
#   permissive - SELinux prints warnings instead of enforcing.
#   disabled - No SELinux policy is loaded.
# See also:
# https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/9/html/using_selinux/changing-selinux-states-and-modes_using-selinux#changing-selinux-modes-at-boot-time_changing-selinux-states-and-modes
#
# NOTE: Up to RHEL 8 release included, SELINUX=disabled would also
# fully disable SELinux during boot. If you need a system with SELinux
# fully disabled instead of SELinux running with no policy loaded, you
# need to pass selinux=0 to the kernel command line. You can use grubby
# to persistently set the bootloader to boot with selinux=0:
#
#   grubby --update-kernel ALL --args selinux=0
#
# To revert back to SELinux enabled:
#
#   grubby --update-kernel ALL --remove-args selinux
#
SELINUX=enforcing
# SELINUXTYPE= can take one of these three values:
#   targeted - Targeted processes are protected,
#   mls - Multi Level Security protection.
SELINUXTYPE=targeted

1,0-1 All
```

3. Review the security context of a file

```
root@localhost:/home/adones

[root@localhost adones]# ls -Z
unconfined_u:object_r:user_home_t:s0 cis-report.html
unconfined_u:object_r:user_home_t:s0 cis-results.xml
unconfined_u:object_r:user_home_t:s0 Desktop
unconfined_u:object_r:user_home_t:s0 Documents
unconfined_u:object_r:user_home_t:s0 Downloads
unconfined_u:object_r:audio_home_t:s0 Music
unconfined_u:object_r:user_home_t:s0 Pictures
unconfined_u:object_r:user_home_t:s0 Public
unconfined_u:object_r:user_home_t:s0 SCC
unconfined_u:object_r:user_home_t:s0 Templates
unconfined_u:object_r:user_home_t:s0 Videos
[root@localhost adones]# ls -laZ /var/lib/
total 16
drwxr-xr-x. 58 root      root      system_u:object_r:var_lib_t:s0      4096 May  3 21:04 .
drwxr-xr-x. 20 root      root      system_u:object_r:var_t:s0         4096 Feb 18 20:16 ..
drwxr-xr-x.  4 root      root      system_u:object_r:accountsdb_var_lib_t:s0 32 Feb 18 20:14 AccountsService
drwx-----.  2 root      root      system_u:object_r:aide_db_t:s0       6 Nov  6 02:45 aide
drwxr-xr-x.  2 root      root      system_u:object_r:alsa_var_lib_t:s0   26 May  3 20:34 alsa
drwxr-xr-x.  2 root      root      system_u:object_r:rpm_var_lib_t:s0   4096 May  3 21:04 alternatives
drwxr-xr-x.  3 root      root      system_u:object_r:var_lib_t:s0      4096 May  3 21:04 authselect
drwxr-xr-x.  3 root      root      system_u:object_r:bluetooth_var_lib_t:s0 18 Feb 18 20:14 bluetooth
drwxr-xr-x.  2 chrony    chrony    system_u:object_r:chronyd_var_lib_t:s0 19 May  3 21:30 chrony
drwxr-xr-x.  3 root      root      system_u:object_r:container_var_lib_t:s0 22 Feb 18 20:20 cni
drwxr-xr-x.  3 root      root      system_u:object_r:colord_var_lib_t:s0 17 Feb 18 20:14 color
drwxr-xr-x.  4 colord    colord    system_u:object_r:colord_var_lib_t:s0 67 Feb 18 20:20 colord
drwxr-xr-x.  4 root      root      system_u:object_r:container_var_lib_t:s0 37 Feb 18 20:20 containers
drwxr-xr-x.  2 root      root      system_u:object_r:system_dbusd_var_lib_t:s0 24 Feb 18 20:17 dbus
drwxr-xr-x.  3 root      root      system_u:object_r:rpm_var_lib_t:s0   93 May  3 21:04 dnf
drwxr-xr-x.  2 root      root      system_u:object_r:dnsmasq_leas_t:s0   6 Sep 16 2025 dnsmasq
drwxrwx---.  2 root      root      system_u:object_r:fapolicyd_var_lib_t:s0 52 May  3 21:02 fapolicyd
drwxr-xr-x.  3 root      root      system_u:object_r:var_lib_t:s0      34 Feb 18 20:22 flatpak
drwx-----.  2 root      root      system_u:object_r:fprintd_var_lib_t:s0 6 Nov 18 16:48 fprint
drwxr-xr-x.  6 root      root      system_u:object_r:fwupd_var_lib_t:s0  84 Feb 18 20:22 fwupd
drwxr-xr-x.  2 root      root      system_u:object_r:games_data_t:s0     6 Nov  2 2024 games
drwxrwx--T.  5 gdm       gdm       system_u:object_r:xdm_var_lib_t:s0   49 Feb 18 20:20 gdm
drwxr-xr-x.  2 geoclue   geoclue   system_u:object_r:geoclue_var_lib_t:s0 6 May 13 2025 geoclue
drwxr-xr-x.  2 root      root      system_u:object_r:hypervkvp_var_lib_t:s0 6 Nov  2 18:26 hyperv
drwxr-xr-x.  2 root      root      system_u:object_r:var_lib_t:s0       6 Dec 17 11:55 initramfs
drwx-----.  3 root      root      system_u:object_r:var_lib_t:s0      17 May  3 21:00 ipsec
drwxr-xr-x.  8 root      root      system_u:object_r:iscsi_var_lib_t:s0  90 Feb 18 20:15 iscsi
```

root@localhost:/home/adones									
drwxr-xr-x.	2	geoclue	geoclue	system_u:object_r:geoclue_var_lib_t:s0	6	May	13	2025	geoclue
drwxr-xr-x.	2	root	root	system_u:object_r:hypervkp_var_lib_t:s0	6	Nov	2	18:26	hyperv
drwxr-xr-x.	2	root	root	system_u:object_r:var_lib_t:s0	6	Dec	17	11:55	initramfs
drwx-----.	3	root	root	system_u:object_r:var_lib_t:s0	17	May	3	21:00	ipsec
drwxr-xr-x.	8	root	root	system_u:object_r:iscsi_var_lib_t:s0	90	Feb	18	20:15	iscsi
drwxr-xr-x.	2	root	root	system_u:object_r:kdump_var_lib_t:s0	6	Oct	31	2025	kdump
drwxr-xr-x.	2	root	root	system_u:object_r:kpatch_var_lib_t:s0	6	Nov	11	08:53	kpatch
drwxr-xr-x.	2	root	root	system_u:object_r:logrotate_var_lib_t:s0	30	May	3	20:22	logrotate
drwxr-xr-x.	2	root	root	system_u:object_r:var_lib_t:s0	6	Nov	2	2024	misc
drwxr-xr-x.	2	root	slocate	system_u:object_r:locate_var_lib_t:s0	24	May	3	20:22	mlocate
drwxr-xr-x.	4	root	root	system_u:object_r:snmpd_var_lib_t:s0	45	Feb	18	20:13	net-snmp
drwx-----.	2	root	root	system_u:object_r:NetworkManager_var_lib_t:s0	171	May	3	21:01	NetworkManager
drwxr-xr-x.	2	root	root	system_u:object_r:bootloader_var_lib_t:s0	6	Feb	4	2025	os-prober
drwxr-xr-x.	2	root	root	system_u:object_r:rpm_var_lib_t:s0	29	May	3	20:35	PackageKit
drwxr-xr-x.	2	root	root	system_u:object_r:plymouthd_var_lib_t:s0	27	Feb	18	20:22	plymouth
drwxr-xr-x.	3	root	polkitd	system_u:object_r:policykit_var_lib_t:s0	28	Feb	18	20:14	polkit-1
drwx-----.	2	postfix	root	system_u:object_r:postfix_data_t:s0	6	Nov	5	2024	postfix
drwxr-xr-x.	2	root	root	system_u:object_r:powerprofiles_var_lib_t:s0	6	Nov	4	2024	power-profiles-daemon
drwx-----.	2	root	root	system_u:object_r:var_lib_t:s0	6	Feb	18	19:52	private
drwxr-xr-x.	2	root	root	system_u:object_r:rasdaemon_var_lib_t:s0	6	May	13	2025	rasdaemon
drwxr-xr-x.	6	root	root	system_u:object_r:rhsmcertd_var_lib_t:s0	91	May	3	20:59	rhsm
drwxr-xr-x.	2	root	root	system_u:object_r:rpm_var_lib_t:s0	91	Oct	30	2025	rpm
drwxr-xr-x.	3	root	root	system_u:object_r:var_lib_t:s0	20	May	3	21:02	rpm-state
drwx-----.	2	root	root	system_u:object_r:syslogd_var_lib_t:s0	29	May	3	21:43	rsyslog
drwxr-xr-x.	5	root	root	system_u:object_r:samba_var_t:s0	46	Feb	18	19:52	samba
drwxr-xr-x.	5	root	root	system_u:object_r:semanage_store_t:s0	46	Feb	18	19:52	selinux
drwx-----.	2	setroubleshoot	setroubleshoot	system_u:object_r:setroubleshoot_var_lib_t:s0	71	Feb	18	20:20	setroubleshoot
drwxr-xr-x.	2	root	root	system_u:object_r:fsdaemon_var_lib_t:s0	6	Apr	20	2024	smartmontools
drwxr-xr-x.	10	root	root	system_u:object_r:sssd_var_lib_t:s0	120	Feb	18	20:15	sss
drwxr-xr-x.	7	root	root	system_u:object_r:init_var_lib_t:s0	98	Dec	3	08:03	systemd
drwxr-xr-x.	3	root	root	system_u:object_r:var_lib_t:s0	20	Feb	18	19:52	tpm2-tss
drwx-----.	2	root	root	system_u:object_r:devicekit_var_lib_t:s0	6	Nov	6	08:50	udisks2
drwxr-xr-x.	2	unbound	unbound	system_u:object_r:named_cache_t:s0	22	Mar	9	22:35	unbound
drwxr-xr-x.	2	root	root	system_u:object_r:devicekit_var_lib_t:s0	127	May	3	21:34	upower
drwxr-xr-x.	3	root	root	system_u:object_r:var_lib_t:s0	23	Feb	18	20:17	xfsdump
drwxr-xr-x.	2	root	root	system_u:object_r:xkb_var_lib_t:s0	29	Feb	18	20:14	xkb

4. Review Policy Booleans

```

root@localhost:/home/adones

virt_use_execmem --> off
virt_use_fusefs --> off
virt_use_glusterd --> off
virt_use_nfs --> on
virt_use_pcsd --> off
virt_use_pulseaudio --> off
virt_use_rawip --> off
virt_use_samba --> off
virt_use_sanlock --> off
virt_use_usb --> on
virt_use_xserver --> off
webadm_manage_user_files --> off
webadm_read_user_files --> off
wine_mmap_zero_ignore --> off
xdm_bind_vnc_tcp_port --> off
xdm_exec_bootloader --> off
xdm_manage_bootloader --> on
xdm_sysadm_login --> off
xdm_write_home --> off
xen_use_nfs --> off
xend_run_blkmap --> on
xend_run_qemu --> on
xguest_connect_network --> on
xguest_exec_content --> on
xguest_mount_media --> on
xguest_use_bluetooth --> on
xserver_clients_write_xshm --> off
xserver_execmem --> off
xserver_object_manager --> off
zabbix_can_network --> off
zabbix_run_sudo --> off
zarafa_setrlimit --> off
zebra_write_config --> off
zoneminder_anon_write --> off
zoneminder_run_sudo --> off
[root@localhost adones]#

```



root@localhost:/home/adones

```
httpd_can_network_connect_cobbler --> off
httpd_can_network_connect_db --> off
httpd_can_network_memcache --> off
httpd_can_network_relay --> off
httpd_can_sendmail --> off
httpd_dbus_avahi --> off
httpd_dbus_sssd --> off
httpd_dontaudit_search_dirs --> off
httpd_enable_cgi --> on
httpd_enable_ftp_server --> off
httpd_enable_homedirs --> off
httpd_execmem --> off
httpd_graceful_shutdown --> off
httpd_manage_ipa --> off
httpd_mod_auth_ntlm_winbind --> off
httpd_mod_auth_pam --> off
httpd_read_user_content --> off
httpd_run_ipa --> off
httpd_run_preupgrade --> off
httpd_run_stickshift --> off
httpd_serve_cobbler_files --> off
httpd_setrlimit --> off
httpd_ssi_exec --> off
httpd_sys_script_anon_write --> off
httpd_tmp_exec --> off
httpd_tty_comm --> off
httpd_unified --> off
httpd_use_cifs --> off
httpd_use_fusefs --> off
httpd_use_gpg --> off
httpd_use_nfs --> off
httpd_use_opencryptoki --> off
httpd_use_openstack --> off
httpd_use_sasl --> off
httpd_verify_dns --> off
```

```
root@localhost:/home/adones

httpd_can_connect_mythtv --> off
httpd_can_connect_zabbix --> off
httpd_can_manage_courier_spool --> off
httpd_can_network_connect --> off
httpd_can_network_connect_cobbler --> off
httpd_can_network_connect_db --> off
httpd_can_network_memcache --> off
httpd_can_network_relay --> off
httpd_can_sendmail --> off
httpd_dbus_avahi --> off
httpd_dbus_sssd --> off
httpd_dontaudit_search_dirs --> off
httpd_enable_cgi --> on
httpd_enable_ftp_server --> off
httpd_enable_homedirs --> off
httpd_execmem --> off
httpd_graceful_shutdown --> off
httpd_manage_ipa --> off
httpd_mod_auth_ntlm_winbind --> off
httpd_mod_auth_pam --> off
httpd_read_user_content --> off
httpd_run_ipa --> off
httpd_run_preupgrade --> off
httpd_run_stickshift --> off
httpd_serve_cobbler_files --> off
httpd_setrlimit --> off
httpd_ssi_exec --> off
httpd_sys_script_anon_write --> off
httpd_tmp_exec --> off
httpd_tty_comm --> off
httpd_unified --> off
httpd_use_cifs --> on
httpd_use_fusefs --> off
httpd_use_gpg --> off
httpd_use_nfs --> off
httpd_use_opencryptoki --> off
```

5. Create a report with screenshots of your results. Include information about your experience using SELinux.

In this assignment, I learned how SELinux works and how it helps secure a system by controlling access using labels and policies. From my screenshots, I confirmed that SELinux was enabled and enforced using commands like `sestatus` and `getenforce`. I also learned how to view the security context of files and directories using `ls -Z`, which helped me understand how different files are labeled. One thing I found interesting was learning how to change the context of a file using the `chcon` command, and that this change is temporary and can go back after a reboot unless it is made permanent using `semanage`. I also learned about policy booleans and how to view and temporarily change them using `setsebool`, which allows

certain permissions to be turned on or off without changing the whole policy. Overall, this assignment helped me better understand how SELinux works and how it is used to improve system security